Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2025-II-Q05] | Part A (Q1-Q5)

Answer: 336

1. The midpoint circle is the nine-point circle; G, H, J are altitude feet. The medial-triangle angles give $\widehat{DE} = 2\angle DFE = 72^\circ$ and $\widehat{FG} = 2\angle FDG = 72^\circ$.

2. H, J also lie on the circle with diameter BC, centered at D. Isosceles triangles BDJ, CDH give $\angle JDB = 60^\circ, \angle HDC = 108^\circ$. Thus $\angle JDH = 12^\circ$ and, as an inscribed angle of the nine-point circle, it gives $\widehat{HJ} = 24^\circ$.

3. The requested sum is $72 + 2 \cdot 24 + 3 \cdot 72 = 336$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2001-II-Q06] | Part B (Q6-Q10)

Answer: 251

1. Let the original side be s, put its center at zero and CD on $y = s/2$. Equal-height points G, H on the circle force horizontal symmetry. For the smaller square of side t, its circle point is $(t/2, s/2 + t)$ outside the original square.

2. The circle equation gives $t^2/4 + (s/2 + t)^2 = s^2/2$, equivalent to $(5t - s)(t + s) = 0$. Hence $t = s/5$. The inward choice gives $t = s$, excluded by the strict area ratio.

3. The area ratio is $1/25$, giving $10 \cdot 25 + 1 = 251$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2011-II-Q10] | Part B (Q6-Q10)

Answer: 057

1. Let the chord midpoints be M, N. Right triangles give $OM = 20, ON = 24$, while $MN = 12$. The angles OMP, ONP are right, placing all four points on the circle with diameter OP.

2. In triangle OMN, the cosine of the included angle is $(400 + 576 - 144)/960 = 13/15$, hence its sine is $2\sqrt{14}/15$.

3. The extended sine rule in the new circle gives $OP = MN/\sin\angle MON = 90/\sqrt{14}$, so $OP^2 = 4050/7$.

4. The numerator-denominator sum is 4057, leaving remainder 57.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2013-II-Q08] | Part B (Q6-Q10)

Answer: 272

1. Let the minor central angles for chords twenty-two and twenty be α, β. No edge can use its major arc: replacing a repeated edge angle by $2\pi - \alpha$ or $2\pi - \beta$ would already force the sum of all six positive arcs to exceed 2π. Thus $4\alpha + 2\beta = 2\pi$.

2. Then $\cos\alpha = \sin(\beta/2) = 10/R$. The chord formula also gives $\cos\alpha = 1 - 242/R^2$.

3. Equating gives $R^2 - 10R - 242 = 0$, whose positive root is $5 + \sqrt{267}$. Thus $p + q = 272$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2015-I-Q06] | Part B (Q6-Q10)

Answer: 058

1. Let each central interval in the second circle be t. Since $\angle ACE = 5t$, the first circle arc AE avoiding C has measure $10t$. Each of its four short intervals is therefore $q = (360 - 10t)/4$.

2. Inscribed-angle rules give $\angle ABD = (360 - 3q)/2$ and $\angle AHG = (360 - 3t)/2$. Their difference 12 simplifies to $21t/4 - 135 = 12$, hence $t = 28$ and $q = 20$ degrees.

3. Then $\angle BAE = 3q/2 = 30^\circ$ and $\angle EAG = t = 28^\circ$. The two minor-arc orders put AB and AG on opposite sides of ray AE, so $\angle BAG = 30 + 28 = 58^\circ$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2016-II-Q10] | Part B (Q6-Q10)

Answer: 043

1. The intersecting-chord relation at Q gives $QC \cdot QT = QA \cdot QB = 42$. Extend AB past B to R with $BR = 8$, so $QP \cdot QR = 3 \cdot 14 = 42$. The converse chord-product theorem shows C, P, T, R are cyclic.

2. Circle angles now give $BRT = PCT = SAT$, while the original cyclic quadrilateral gives the exterior angle $RBT = AST$. Thus triangles RBT, AST are similar.

3. The corresponding lengths satisfy $ST/BT = AS/RB = 7/8$. Hence $ST = 35/8$ and $m + n = 43$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2023-II-Q09] | Part B (Q6-Q10)

Answer: 033

1. Tangency parallel to a chord places the tangency point over that chord midpoint. Consequently, taking $P=(0,0), X=(-10,0), Y=(14,0)$ gives $A=(-5,h), B=(7,h)$ and $AB=12, XY=24$.

2. The radical axis PQ meets AB at its midpoint $M=(1,h)$, since the powers MA^2, MB^2 agree. Reflection in the line of the two circle centers interchanges P, Q and the two common external tangents. Thus choosing the tangent nearer P makes its distance from P smaller than its distance from Q. Since M, P, Q are collinear and M has positive circle power, their order is M, P, Q, so $MP \cdot MQ = MA^2 = 36$ and $MQ = MP + 5$.

3. Thus $MP = 4$, and $h^2 = MP^2 - 1^2 = 15$. The alternative same-side chord-end placement would give $AB = 2, |M_x| = 6$; then $MP \geq 6$ and $MP \cdot MQ \geq 6(6-5) > 1 = MA^2$, impossible.

4. The area is $(24+12)\sqrt{15}/2 = 18\sqrt{15}$, yielding $m+n=33$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2024-I-Q10] | Part B (Q6-Q10)

Answer: 113

1. The tangent–chord theorem makes DBC isosceles, with base angles A. Cosine rule gives $\cos A = 11/25$, so $DB = (9/2)/(11/25) = 225/22$.

2. Also $\cos B = 1/15$, and $\cos(A+B) = -13/15$. Applying cosine rule in ABD yields $DA^2 = 25 + (225/22)^2 + 2 \cdot 5(225/22)(13/15) = (325/22)^2$.

3. Power of D gives $DP \cdot DA = DB^2$. Thus $AP = DA - DP = (DA^2 - DB^2)/DA = 100/13$ and $m+n = 113$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2025-I-Q06] | Part B (Q6-Q10)

Answer: 504

1. The height is the incircle diameter, 6, so the area gives $r+s=24$. Equal tangent lengths imply the two legs also sum to 24, hence each leg is 12.

2. An end right triangle gives $((r-s)/2)^2=12^2-6^2=108$. Therefore $r^2+s^2=[(r+s)^2+(r-s)^2]/2=(576+432)/2=504$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2026-II-Q10] | Part B (Q6-Q10)

Answer: 340

1. Angle-bisector ratios give $BD : CD = 8 : 9$. As both are integers, write $BD = 8t, CD = 9t$ with positive integer t. Powers of B, C yield $BE = 8t^2/25, CF = 9t^2/25$.

2. Integrality requires $5 \mid t$, say $t = 5s$. Then $AE = 200 - 8s^2 > 0$ and $AF = 225 - 9s^2 > 0$, so $s \leq 4$. Hence $BC = 85s \leq 340$.

3. For $s = 4$, the side lengths $200, 225, 340$ form a triangle and give $BD = 160, CD = 180, AE = 72, AF = 81$, all positive integers. The maximum is 340.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2001-I-Q13] | Part C (Q11-Q15)

Answer: 174

1. Set $t = d/2$ and $c = \cos t$. Chord ratios give the $2d$ and $3d$ lengths as $44c$ and $22(4c^2 - 1)$.

2. Their difference condition reduces to $44c^2 - 22c - 1 = 0$. As $t < 60^\circ$, take $c = (11 + \sqrt{165})/44$.

3. The required chord equals $44c - 20 = -9 + \sqrt{165}$, giving 174.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2002-I-Q13] | Part C (Q11-Q15)

Answer: 063

1. Let P be the centroid. Then $AE = 12$, $AP = 12$, $PE = 9$. The chord power at E gives $EC \cdot EF = EA \cdot EB = 144$, so $EF = 16/3$.

2. In isosceles AEP with equal sides 12 and base 9, the height to PE is $3\sqrt{55}/2$. Hence $[AEP] = 27\sqrt{55}/4$.

3. Collinear-base scaling gives $[AEF] = (EF/EP)[AEP] = 4\sqrt{55}$. Since E bisects AB, $[AFB] = 8\sqrt{55}$, so the answer is 63.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2005-I-Q15] | Part C (Q11-Q15)

Answer: 038

1. Let M be the midpoint of AB, let R be the incircle contact point on AC, and let the incircle cut CM at one-third and two-thirds positions. Its tangent lengths from M and C have equal square $2CM^2/9$. Relabel A, B so the base tangency point T lies in AM. Equal tangent lengths give $AC = AR + CR = AT + MT = AM = 10$.

2. Put $a = BC$. With semiperimeter $15 + a/2$, $MT = (a - 10)/2$ and median formula gives $CM^2 = (a^2 - 100)/2$. Thus $(a - 10)^2/4 = (a^2 - 100)/9$. Since $a > 10$, cancellation yields $9(a - 10) = 4(a + 10)$, so $a = 26$.

3. Heron gives area $\sqrt{28 \cdot 8 \cdot 18 \cdot 2} = 24\sqrt{14}$. The requested sum is 38.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2009-I-Q15] | Part C (Q11-Q15)

Answer: 150

1. Cosine rule gives $\angle A = 60^\circ$. The incenter-angle identities give $\angle BI_BD + \angle CI_CD = 180^\circ + \angle A/2 = 210^\circ$. Cyclic-angle relations put the second intersection below BC and yield $\angle BPC = 150^\circ$, independent of D.

2. Thus P lies on a fixed circular arc through B, C. For fixed chord, maximal perpendicular distance occurs at the arc midpoint, where $BP = PC = x$. This point is attainable: as D varies continuously from B to C, its first circle collapses to B at one end and its second to C at the other, so the distinct second intersection varies continuously from B to C along that arc and must pass its midpoint.

3. Cosine rule at the midpoint gives $196 = (2+\sqrt{3})x^2$. Hence the maximal area is $x^2 \sin 150^\circ/2 = 49(2-\sqrt{3}) = 98 - 49\sqrt{3}$, and the requested sum is 150.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2010-II-Q15] | Part C (Q11-Q15)

Answer: 218

1. Angle bisectors give $AD = 195/29$, $AE = 65/9$. Thus $AD - AM = 13/58$ and $AN - AE = 5/18$.

2. Use unit vectors u, v from A toward C, B and write points as $su + tv$. A circle through A, meeting these rays at distances d, e, has equation $s^2 + t^2 + 2st\cos A - ds - et = 0$.

3. Subtract the equations of the two circles. Their common chord AP has $(AD - AM)s + (AE - AN)t = 0$, so $t/s = (13/58)/(5/18) = 117/145$.

4. On BC, oblique coordinates give $BQ/CQ = (15/13)(s/t) = 725/507$. The factors $725 = 5^2 \cdot 29$ and $507 = 3 \cdot 13^2$ are coprime, so $m - n = 218$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2012-II-Q15] | Part C (Q11-Q15)

Answer: 919

1. Let E' be antipodal to E on ω. Both $\angle DFE$ and $\angle E'FE$ are right, so E', D, F are collinear. This reduces the second-circle intersection to a line-circle intersection.

2. Take $B = (0,0), C = (7,0), A = (65/14, 15\sqrt{3}/14)$. The bisector gives $D = (35/8, 0)$. The circumcenter is $(7/2, -7\sqrt{3}/6)$ with radius squared $49/3$. The second bisector-circle intersection is the midpoint of the arc BC not containing A: it lies on $x = 7/2$ and on ray AD beyond D, hence $E = (7/2, -7\sqrt{3}/2)$. Its antipode is therefore $E' = (7/2, 7\sqrt{3}/6)$.

3. Parameterize line $E'D$ by $E' + t(3, -4\sqrt{3})$. Substitution in the circle equation gives $57t^2 - 56t = 0$. The second point is $t = 56/57$, yielding $F = (245/38, -105\sqrt{3}/38)$.

4. Therefore $AF^2 = (240/133)^2 + 3(510/133)^2 = 900/19$. It is reduced, so $m + n = 919$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2014-I-Q15] | Part C (Q11-Q15)

Answer: 041

1. Take $B=(0,0), A=(0,3), C=(4,0), E=(0,e), D=(d,0)$. Since angle EBD is right, ED is a circle diameter and its equation is $x^2+y^2=dx+ey$. The point equidistant from E, D on the circle and in the positive quadrant is $F=((d+e)/2,(d+e)/2)$. As AC is $3x+4y=12$, this forces $d+e=24/7$.

2. Substitute $y=(12-3x)/4, d=24/7-e$ in the circle equation and remove the known root $x_F=12/7$. The second point is $G=(x,y)=(28(3-e)/25,(12+21e)/25)$. These coordinates satisfy $G-D=(-y/7,y)$ and $G-E=(x,x/7)$.

3. Therefore $DG/EG=y/x=3/4$. Together with $3x+4y=12$, this gives $G=(2,3/2)$. The circle equation now says $2d+(3/2)e=25/4$.

4. Combine with $d+e=24/7$ to obtain $d=31/14, e=17/14$, both within the sides. Thus $DE=\sqrt{d^2+e^2}=25\sqrt{2}/14$, and the sum is 41.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2016-I-Q15] | Part C (Q11-Q15)

Answer: 270

1. Let M be the midpoint of the common tangent segment AB. Its powers to the two circles are $MA^2 = MB^2$, so it lies on their radical axis XY. Both circles, and hence X, Y, lie in the same half-plane of AB. Their near/far order is therefore M, X, Y, with $MX \cdot MY = MA^2$.

2. All angles in the next two steps are directed modulo 180°, so replacing a ray by its opposite does not affect the statements. Cyclicity gives $\angle(DAX) = \angle(DYX)$ and $\angle(CBX) = \angle(CYX)$. Since D, Y, C are collinear, these angles agree. If $AD \parallel BC$, this would force $AX \parallel BX$, contradicting $X \notin AB$. Thus $Z = AD \cap BC$ exists, and the radical-axis theorem places it on XY.

3. The same angle equality now reads $\angle ZAX = \angle ZBX$, proving A, B, X, Z concyclic. From this circle and the tangent-chord theorem on the second original circle, $\angle AZX = \angle ABX = \angle BYX$. Since $ZX \parallel YX$, it follows that $ZA \parallel BY$. Interchanging the two original circles similarly gives $ZB \parallel AY$.

4. Thus $AYBZ$ is a parallelogram. Its diagonals AB, YZ bisect each other at M, so Z is the reflection of Y through M. The full order on their common line is Z, M, X, Y. In particular $XZ = MX + MY$ and $XY = MY - MX$.

5. For the similarity, Z, C, B are collinear and Z, D, A are collinear. Cyclicity of B, C, Y, X gives $\angle XCZ = \angle XYB = \angle XZD$, using $YB \parallel ZA$. Likewise $\angle XZC = \angle XYA = \angle XDZ$, using $ZC \parallel YA$ and cyclicity of A, D, Y, X. Therefore triangles XZC and XDZ are similar, and $XZ/XD = XC/XZ$, hence $XZ^2 = XC \cdot XD$.

6. Finally $AB^2 = 4MX \cdot MY = (MX + MY)^2 - (MY - MX)^2 = XZ^2 - XY^2 = 67 \cdot 37 - 47^2 = 270$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2018-I-Q15] | Part C (Q11-Q15)

Answer: 059

1. Write one cyclic order of side arcs as $\alpha, \beta, \gamma, \delta$, totaling 2π. The other two orders exchange the three possible opposite-side pairings. By the intersecting-chord angle theorem, their diagonal-angle sines are the sines of half the corresponding opposite-arc sums.

2. The diagonals in any one quadrilateral have lengths twice the diagonal-angle sines of the other two orders, using the unit-circle chord formula. Consequently the common area is $\frac{1}{2}d_1 d_2 \sin\phi = 2\sin\phi_A \sin\phi_B \sin\phi_C$. This formula is unchanged if an angle is replaced by its supplement.

3. Substitute to obtain $K = 2(2/3)(3/5)(6/7) = 24/35$, hence $m + n = 59$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2018-II-Q14] | Part C (Q11-Q15)

Answer: 227

1. Let the tangencies on AB, AC be Z, W and their common distance from A be ℓ. Because Y is the first intersection from A, the tangent cuts off the corner: $PY = PZ = \ell - AP$ and $BX = BZ = AB - \ell$.

2. The tangent-chord angle theorem makes the angles opposite AP in triangle APY and opposite AB in triangle ABX supplementary. Their angle at A is common. The sine rule therefore gives $PY/AP = BX/AB$, or $\ell/AP + \ell/AB = 2$.

3. With $AP = 3$, $AB = 7$, this yields $\ell = 21/5$. The same argument on AC gives $\ell/AQ + \ell/AC = 2$, hence $1/AQ = 10/21 - 1/8 = 59/168$.

4. Thus $AQ = 168/59$, within AC, giving 227.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2019-I-Q13] | Part C (Q11-Q15)

Answer: 032

1. The cyclic angles give $\cos\angle DFE = \pm\cos\angle ACB = \pm 3/4$. With $DF = 2, EF = 7$, the two possible values of $\delta = DE$ are $\sqrt{32}$ and $\sqrt{74}$; the point order must select the valid one.

2. Use $A = (0,0), B = (4,0), C = (27/8, 15\sqrt{7}/8)$, $D = (d,0), E = (d+\delta, 0)$, and $F = (d+w, y)$. Distance subtraction gives $w = (\delta^2 - 45)/(2\delta)$ and $y^2 = 4 - w^2$. The two circle equations are $x^2 + y^2 - dx + (27d - 288)y/(15\sqrt{7}) = 0$ and $x^2 + y^2 - (d+\delta+4)x - (180 + 5d + 5\delta)y/(15\sqrt{7}) + 4(d+\delta) = 0$.

3. Substitute the two signs of y into both equations. For $\delta = \sqrt{32}$ the common solution has $y = -7\sqrt{14}/16$ and $d = (21 + 5\sqrt{2})/4 > 4$. For $\delta = \sqrt{74}$ it has $y = 7\sqrt{518}/148$ and $d = (84 - 5\sqrt{74})/26 < 4$, violating the required order. The opposite sign in each case does not satisfy both circles. Thus only the first branch is allowed.

4. It gives $BE = d + \delta - 4 = (5 + 21\sqrt{2})/4$. The explicitly obtained coordinates satisfy both circle equations and both distances, so the branch is attainable. The requested sum is $5 + 21 + 2 + 4 = 32$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2019-I-Q15] | Part C (Q11-Q15)

Answer: 065

1. The inner circles touch ω at A, B. Let the tangents there meet at Z. Equal tangent lengths make its powers in the two inner circles equal, so Z lies on PQ. With $s = ZP$, midpoint M of AB gives $ZA^2 - s^2 = MA^2 - MP^2 = 16 - 1 = 15$.

2. Let H be midpoint of XY and $t = ZH$. Power at Z in ω gives $t^2 - (11/2)^2 = s^2 + 15$, so $t^2 - s^2 = 181/4$. Power at P gives $PH^2 = (11/2)^2 - AP \cdot PB = 61/4$. As $t > s$, $PH = t - s$.

3. Subtracting yields $2s(t - s) = 30$, hence $t = s + 15/s$. Power in an inner circle gives $ZQ = ZA^2/ZP = s + 15/s = t$, so $Q = H$. Therefore $PQ^2 = 61/4$, giving 65.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2019-II-Q15] | Part C (Q11-Q15)

Answer: 574

1. Write $b = AC, c = AB, a = BC, k = \cos A > 0$, $u = AP = bk, v = AQ = ck$. Similar triangles APQ, ACB give $ak = 25$. Since the feet lie inside the circle and $XP, QY < PQ$, the order is X, P, Q, Y. Powers give $u(c - u) = 400$ and $v(b - v) = 525$.

2. Put $w = bck = uv/k$. Then $u^2 = w - 400, v^2 = w - 525$. Cosine rule multiplied by k^2 gives $625 = u^2 + v^2 - 2uvk$, so $wk^2 = w - 775$.

3. Now $(w - 400)(w - 525) = (uv)^2 = w^2k^2 = w(w - 775)$, yielding $w = 1400, k^2 = 25/56$. Therefore $bc = w/k = 560\sqrt{14}$ and the answer is 574.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2020-II-Q15] | Part C (Q11-Q15)

Answer: 717

1. Tangent-chord and sine-law relations yield $TX = 16\sin C$, $TY = 16\sin B$. Since A, X, T, Y are concyclic with diameter AT, $XY = AT\sin A$. For circumradius R, the given sum is $[64(AB^2 + AC^2) + 121AT^2]/R^2 = 1143$.

2. Set $B = (-11, 0), C = (11, 0), T = (0, -\sqrt{135})$. Perpendicular radius and tangent give circumcenter $O = (0, 121/\sqrt{135})$, hence $R^2 = 30976/135$.

3. For $A = (u, v)$ on the circle, $u^2 + v^2 = 242v/\sqrt{135} + 121$. Therefore $AB^2 + AC^2 = 484v/\sqrt{135} + 484$ and $AT^2 = 512v/\sqrt{135} + 256$.

4. The sum condition gives $v = 291/\sqrt{135}$. Thus $AT^2 = 183552/135$ and $XY^2 = 121AT^2/R^2 = 183552/256 = 717$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2021-I-Q11] | Part C (Q11-Q15)

Answer: 301

1. Let P be the diagonal intersection and θ their acute angle. Reflection across an angle bisector swaps the two diagonal lines; multiplying by $\cos\theta$ then sends every vertex to its foot on the other diagonal. Thus all side lengths scale by $\cos\theta$ and the new perimeter is $22\cos\theta$.

2. Ptolemy gives $AC \cdot BD = 4 \cdot 6 + 5 \cdot 7 = 59$. The area is $\sqrt{7 \cdot 6 \cdot 5 \cdot 4} = 2\sqrt{210}$, so $\sin\theta = 4\sqrt{210}/59$.

3. Therefore $\cos\theta = 11/59$, and the perimeter is $242/59$. The requested sum is 301.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2021-II-Q14] | Part C (Q11-Q15)

Answer: 592

1. Let M be the midpoint of BC. The centroid gives A, G, M collinear. Since $OA \perp AX$ and $OG \perp GX$, points O, A, X, G are cyclic. Also $OM \perp BC$ and $OG \perp GY$, so O, G, Y, M are cyclic.

2. The two cyclic quadrilaterals imply the same directed angle modulo 180°: $\angle(OX, OA) = \angle(GX, GA)$ and $\angle(OY, OM) = \angle(GY, GM)$. The right-hand angles are equal modulo 180° because X, G, Y and A, G, M are collinear. Hence ordinary $\angle XOY$ is either $\angle AOM$ or its supplement; this step alone does not fix the rays.

3. Write $B = 13t, C = 2t, A = 180^\circ - 15t$. Acuteness forces $6 < t < 90/13$. Consequently the given $\angle XOY = 17t$ is obtuse. The ray OM points toward the midpoint of the minor arc BC; the chord/arc-angle relations give $\angle AOM = A + 2C = 180^\circ - 11t$, also obtuse throughout this interval. Therefore the supplementary alternative is excluded.

4. We must have $17t = 180^\circ - 11t$, yielding $t = 45/7$. Thus $A = 585/7$ degrees; all three angles are acute and the indicated nondegenerate intersections exist. The reduced numerator-denominator sum is $585 + 7 = 592$.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2022-I-Q11] | Part C (Q11-Q15)

Answer: 150

1. The tangent lengths from A, C are $\sqrt{3 \cdot 12} = 6$ and $\sqrt{16 \cdot 25} = 20$. Writing $AB = x$, equal tangents at B give $BC = x + 14$.

2. Set $\angle BAD = 2t$ and circle radius r. The tangent length from A gives $r = 6 \tan t$. The distance between the parallel tangent sides is $x \sin 2t = 2r$, so $x \cos^2 t = 6$.

3. The cosine law along $AC = 28$ gives $784 = x^2 + (x + 14)^2 + 2x(x + 14) \cos 2t$. Substitute $\cos 2t = 12/x - 1$ to obtain $784 = 24x + 532$, hence $x = 21/2$ and $\cos^2 t = 4/7$.

4. Then $\sin 2t = 4\sqrt{3}/7$. The area is $(21/2)(49/2)(4\sqrt{3}/7) = 147\sqrt{3}$, so the requested sum is 150.

Source / 来源

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

[AIME-2022-II-Q15] | Part C (Q11-Q15)

Answer: 140

1. Let the tangent-circle radii be r, s, so $r + s = 15$; put $t = rs > 0$. Diagonal O_1O_2 splits the hexagon into cyclic quadrilaterals with side lists $(r, 2, s, 15)$ and $(r, 16, s, 15)$. Brahmagupta gives their squared areas $K_2^2 = 14(t + 16)$ and $K_{16}^2 = 56(t + 184)$.

2. For a cyclic quadrilateral with consecutive sides u, v, w, z, its circumradius satisfies $16K^2R^2 = (uv + wz)(uw + vz)(uz + vw)$. Both pieces share the same R. Substituting the two side lists gives $R^2 = (6750 + 169t)(t + 30)/[224(t + 16)]$ and $R^2 = (54000 + t)(t + 240)/[896(t + 184)]$.

3. Equate these positive fractions and clear denominators. The resulting equation factors as $3(t - 40)(225t^2 + 48136t + 486000) = 0$. The quadratic factor is positive for $t > 0$, so $t = 40$.

4. The areas are therefore $\sqrt{14 \cdot 56} = 28$ and $\sqrt{56 \cdot 224} = 112$. Convexity makes their interiors disjoint on opposite sides of the diagonal, so the required total is 140.

Source / 来源